

# Chapter 3

## Points, Part 1 of 2

**W**e now begin the development of the subject of inversion of the real plane in a circle in earnest and we shall do that in the style of a typical college-level presentation, in which definitions are followed by a chain of theorems accompanied by their proofs. The hierarchy of such theorems thus constructed represents the local corpus of layer-specific knowledge.

We will intersperse the presentation of the relevant theoretical material by a number of light-weight problems and, later on, we will take on more substantive problems. For best results we recommend that our readers try to generate their own proofs of all the theorems in this book before looking at our versions.

**Theorem 1.** Under the inversion of a plane with respect to a circle  $c$  with the center at the point  $O$  and the radius  $r$  with positive power any point  $A$  on the circle of inversion  $c$  is transformed into itself.

**Proof.** For any such point  $A$  on  $c$ , according to **Condition 3** of **Definition 1**, we have:

$$|OA| \times |OA'| = r^2$$

where the point  $A'$  is the image of the point  $A$  under the given inversion.

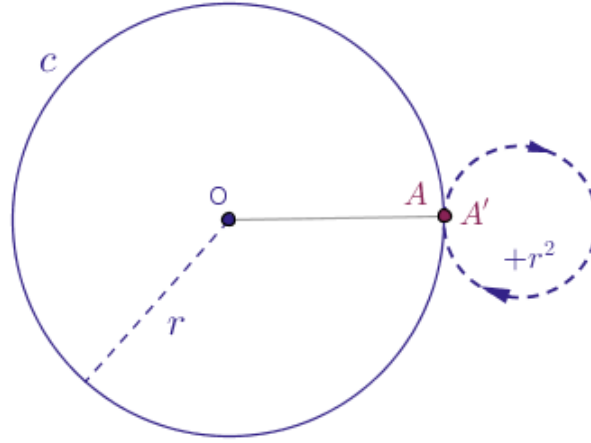
Because the point  $A$  is located on the circle  $c$ , it follows that  $|OA| = r$ . Hence:

$$r \times |OA'| = r^2$$

from where:

$$|OA'| = r$$

According to **Definition 2**, since the power of inversion is positive, the points  $A$  and  $A'$  are located on the same side of the center of inversion  $O$ . Hence, due to **Condition 1** of **Definition 1**, which states the the points  $O$ ,  $A$  and  $A'$  all belong to one and the same straight line, the equality  $|OA'| = r$  means that the image point  $A'$  does indeed coincide with the original point  $A$  (Figure 3.1):



If, for the sake of brevity, we agree to designate a circle,  $c$ , centered at a point  $O$  with the radius  $r$  as  $c(O, r)$  then we just proved that under the inversion of a plane with respect to a circle  $c(O, r)$  with positive power any point  $A$  on  $c$  is taken into itself.  $\square$

Note that such points which, as a result of a transformation, do not move at all or remain stationary are called *fixed*.

---

**Theorem 2:** under the inversion of a plane with respect to a circle  $c(O, r)$  with negative power any point  $A$  on  $c$  is transformed into a point on  $c$  that is diametrically opposite to  $A$ .

**Proof.** For any such point  $A$  on  $c$ , according to **Condition 3** of **Definition 1**, we have:

$$|OA| \times |OA'| = r^2$$

where the point  $A'$  is the image of the point  $A$  under the given inversion.

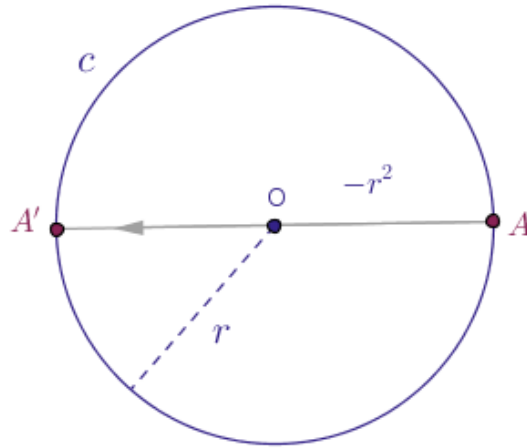
Because the point  $A$  is located on the circle  $c$ , it follows that  $|OA| = r$ . Hence:

$$r \times |OA'| = r^2$$

from where:

$$|OA'| = r$$

According to **Definition 3**, since the power of inversion is negative, the center of inversion  $O$  in this case is located in between the points  $A$  and  $A'$ . Hence, due to **Condition 1** of **Definition 1**, which states the the points  $O$ ,  $A$  and  $A'$  all belong to one and the same straight line, the equality  $|OA'| = r$  in this case means that the points  $A$  and  $A'$  are located on the opposite sides of a diameter of the circle of inversion (Figure 3.2):



Thus, under the inversion of a plane with respect to a circle  $c(O, r)$  with negative power any point  $A$  on  $c$  is transformed into a point on  $c$  that is diametrically opposite to  $A$ .  $\square$

---

**Theorem 3.** Under the inversion of a plane with respect to a circle  $c(O, r)$  the circle of inversion  $c$  is transformed into itself.

**Proof.** The proof of this theorem follows directly from **Theorem 1** and **Theorem 2**.  $\square$

Note that the above theorem applies to inversions of a plane with both negative and positive powers alike with the following wrinkle.

Under the inversion of a plane with respect to a circle with positive power the circle of inversion is transformed into itself in such a way that each and every point of that circle is taken precisely back into itself. Hence, in that case each and every *point* of the circle of inversion is fixed or stationary.

Under the inversion of a plane with respect to a circle with negative power the circle of inversion is also transformed into itself but in a slightly different way because in that case each and every point of the original circle of inversion is taken into its diametrically opposite image which, of course, is still located on the same circle of inversion.

Intuitively, we may, thus, say that under the inversion of a plane with respect to a circle with negative power the points on the circle of inversion trade places. Which means that, while the circle of inversion in this case is literally rebuilt in a non-trivial fashion point-by-point because each point on that circle trades places with its diametrically opposite counterpart, the result of this re-wiring of the circle of inversion comes out to be the original circle nonetheless. This phenomenon is both curious and cool.

In other words, under the inversion of a plane with respect to a circle with negative power none of the points that make that circle of inversion remain fixed or stationary but the resultant shape of that transformation is still the original one!

Either way, such figures which, as a result of a transformation, remain the same are called the *invariant figures* and the corresponding transformations are called the *identity transformations*.

---

**Theorem 4.** under the inversion of a plane with respect to a circle  $c(O, r)$  any point  $A \neq O$  located inside of  $c$  is transformed into an image point  $A'$  that is located outside of  $c$ .

**Proof.** For any such point  $A$  that is located inside of  $c$ , according to **Condition 3** of **Definition 1**, we have:

$$|OA| \times |OA'| = r^2$$

Since the point  $A \neq O$  is located strictly **inside** of the circle of inversion, it follows that  $|OA| < r$ . Hence, from the above equality it follows that:

$$|OA'| = \frac{r^2}{|OA|} > \frac{r^2}{r} = r$$

from where:

$$|OA'| > r$$

which precisely means that under the inversion of a plane with respect to a circle  $c(O, r)$  any point  $A \neq O$  located **inside** of  $c$  is transformed into an image point  $A'$  that is located **outside** of  $c$ .  $\square$

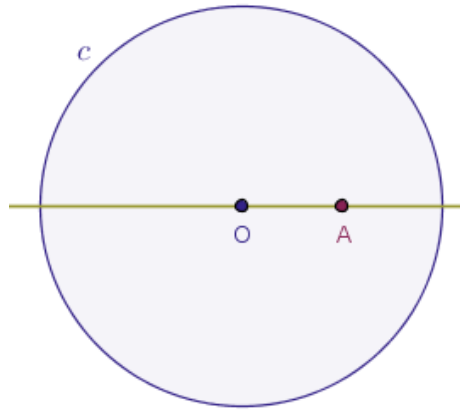
Note again that the above theorem applies to inversions of a plane in a circle with both negative and positive powers alike.

---

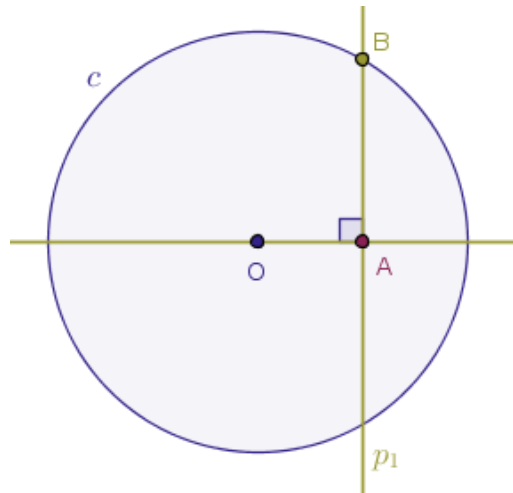
**Problem 1.** Given a circle of inversion  $c(O, r)$  and a point  $A \neq O$  located inside of  $c$ , construct  $A$ 's inverse with respect to  $c$  with positive power using the Euclidean tools, a compass and a straightedge, alone.

**Solution.** Our definition of inversion of a plane with respect to a circle with positive power suggests the following construction steps.

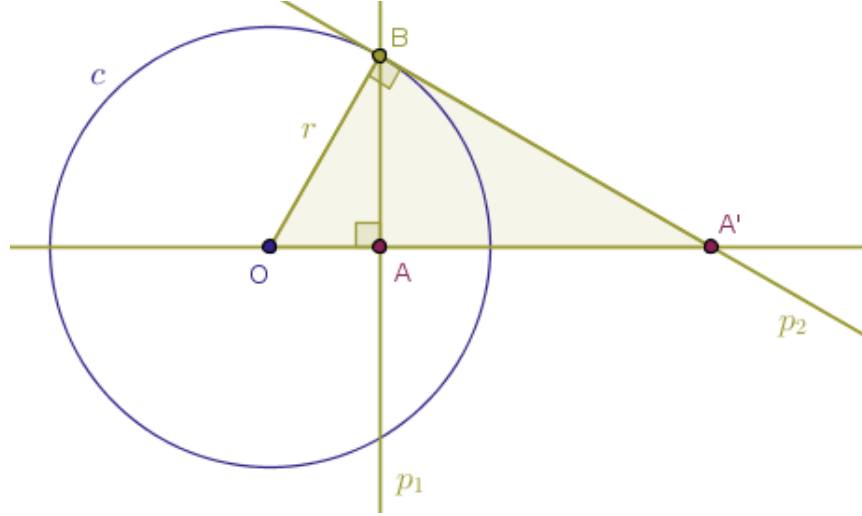
1. Draw a straight line through the points  $O$  and  $A$  (Figure 3.3):



2. Draw a straight line,  $p_1$ , through the point  $A$  perpendicular to the straight line  $OA$  until  $p_1$  intersects the circle  $c$  at a point,  $B$  (Figure 3.4):



3. Draw a straight line through the points  $O$  and  $B$ , draw a straight line,  $p_2$ , through the point  $B$  perpendicular to the straight line  $OB$  until  $p_2$  intersects the straight line  $OA$  extended at the point  $A'$  (Figure 3.5):



The point  $A'$  thus obtained is the sought-after image of the original point  $A$  under the inversion of a plane with respect to the circle  $c$  with positive power.

Proof. By construction the points  $A$  and  $A'$  are located on the same side of the center of inversion  $O$ , these points are collinear with  $O$  and these points are coplanar with the plane containing the circle of inversion  $c$ .

The right, by construction, triangles  $OAB$  and  $OB A'$  are similar due to Euclid's "*Elements*" Book 6 Proposition 8. In such triangles, by the same Book 6 Proposition 4, the lengths of the following sides about the equal angles are in the same proportion:

$$\frac{|OA|}{|OB|} = \frac{|OB|}{|OA'|}$$

from where it follows that:

$$|OB|^2 = r^2 = |OA| \times |OA'|$$

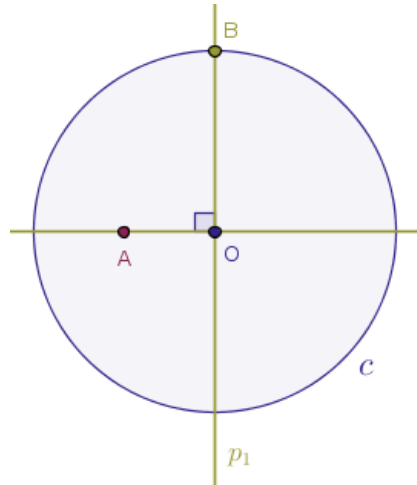
which satisfies **Condition 3** of **Definition 1**.  $\square$

---

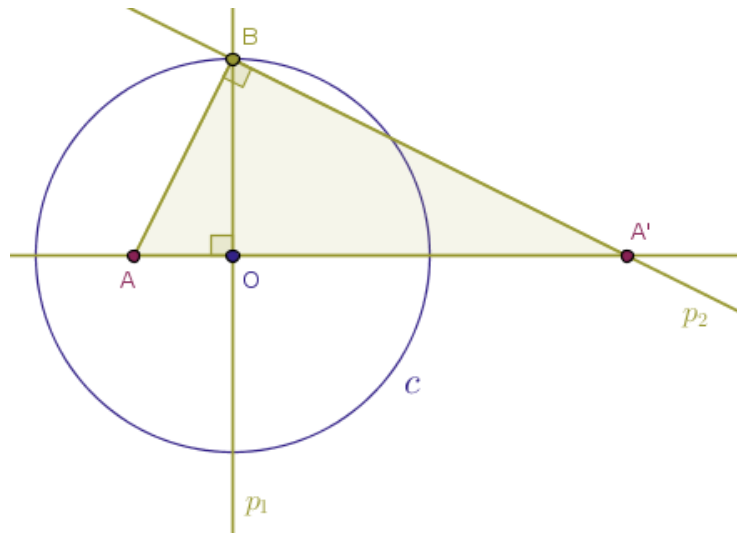
**Problem 2.** Given a circle of inversion  $c(O, r)$  and a point  $A \neq O$  located inside of  $c$ , construct  $A$ 's inverse with respect to  $c$  with negative power using a compass and a straightedge alone.

**Solution.** Our definition of inversion of a plane with respect to a circle with negative power suggests the following construction steps.

1. Draw a straight line through the points  $O$  and  $A$ , draw a straight line,  $p_1$ , through the point  $O$  perpendicular to the straight line  $OA$  until  $p_1$  intersects  $c$  at  $B$  (Figure 3.6):



2. Draw a straight line through the points  $A$  and  $B$ , draw a straight line,  $p_2$ , through the point  $B$  perpendicular to the straight line  $AB$  until  $p_2$  intersects the straight line  $OA$  extended at  $A'$  (Figure 3.7):



The point  $A'$  thus obtained is the sought-after image of the original point  $A$  under the inversion of a plane with respect to the circle  $c$  with negative power.

Proof. By construction the triangle  $ABA'$  is right and the line segment  $BO$  is its height that emanates from the right-angle vertex  $B$  and falls onto the hypotenuse  $AA'$ .

Hence, by Euclid's “*Elements*” Book 6 Proposition 8, the right triangle  $ABA'$  is similar to the right triangles  $AOB$  and  $BOA'$ , which are similar as well.

By the same Book 6 Proposition 4, in equiangular triangles the sides about the equal angles are proportional where the corresponding sides are opposite the equal angles, for the similar right triangles  $AOB$  and  $BOA'$  we have for the sides that form the respective right angles in these triangles:

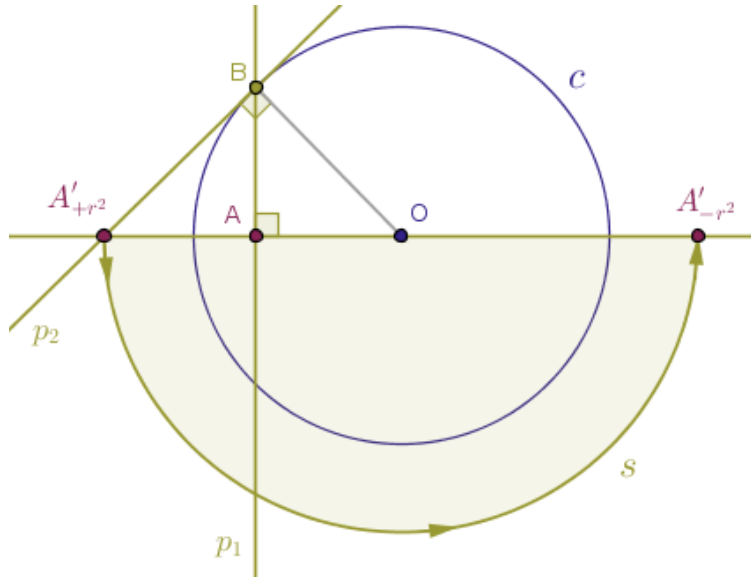
$$\frac{|OA|}{|OB|} = \frac{|OB|}{|OA'|}$$

from where it follows that:

$$|OA| \times |OA'| = |OB|^2 = r^2$$

The points  $O$ ,  $A$  and  $A'$  are collinear and are coplanar with the circle of inversion  $c$ . Hence, the point  $A'$  is indeed the image of the point  $A$  under the inversion of the parent plane with respect to the circle  $c$  with negative power.  $\square$

Observe that an alternative construction of the inverse image of a given point with negative power exists. That is, in order to construct the image of a given point under the inversion of the parent plane with respect to a circle with *negative* power, it is possible to construct the respective image, the point  $A'_{+r^2}$  in the diagram below, with *positive* power, as was explained in **Problem 1**, first (Figure 3.8):



and then rotate the image point  $A'_{+r^2}$  in the parent plane about the center of inversion by  $\pi$  radians in either direction. Such a rotation will put the image point  $A'_{+r^2}$  onto the straight line  $OA$  extended precisely into the point  $A'_{-r^2}$  that is the sought-after image of the point  $A$  under the respective inversion with negative power.



Technically the rotation of the parent plane about the center of inversion  $O$  in this case amounts to constructing a circle,  $s$ , that has  $O$  as its center and the distance  $|OA'_{+r^2}|$  as its radius. That circle will intersect the straight line  $OA$  extended at two points, the point  $A'_{+r^2}$  and the point  $A'_{-r^2}$ , of which we are interested in the point  $A'_{-r^2}$ .

For a proof note that 1) since it is the case that  $|OA| \times |OA'_{+r^2}| = r^2$  by construction and 2)  $|OA'_{+r^2}| = |OA'_{-r^2}|$  by construction as well, it follows that  $|OA| \times |OA'_{-r^2}| = r^2$ , the points  $O$ ,  $A$  and  $A'_{-r^2}$  are collinear and coplanar with the circle of inversion and the center of inversion, the point  $O$ , is in between the preimage point,  $A$ , and the image point,  $A'_{-r^2}$ .

As such, the point  $A'_{-r^2}$  is the inverse of the point  $A$  with negative power and conversely. If one and the same point  $A$  is reflected in the same plane and in the same circle but with negative power into the point  $A'_{-r^2}$  then rotating the point  $A'_{-r^2}$  in the parent plane about the center of inversion,  $O$ , by half-a-turn will put that point precisely onto the point  $A'_{+r^2}$ .

Since the points  $O$  and  $A$  are collinear with:

- the point  $A'_{+r^2}$ , which is the inverse image of  $A$  with positive power and with
- the point  $A'_{-r^2}$ , which is the inverse image of  $A$  with negative power

it follows that the points  $A'_{+r^2}$  and  $A'_{-r^2}$  must be collinear and rotating one of these points in the parent plane about the center of inversion by  $\pi$  radians will take that point into the remaining one.

This proof shows that working the above construction steps in reverse order it is always possible to obtain the image of the point  $A$  with positive power if the corresponding image with negative power is constructed or known first.

To that end, using the steps described in **Problem 2**, construct the image, the point  $A'_{-r^2}$ , of the point  $A$  with *negative* power first and then rotate the point  $A'_{-r^2}$  in the parent plane about the center of inversion by  $\pi$  radians in either direction into the point  $A'_{+r^2}$  on the straight line  $OA$  extended, which will be exactly the inverse image of the point  $A$  with *positive* power.

Overall we just proved that construction-wise the inverse images with positive and negative powers are equivalent in the sense that if the location of one such image is known then the location of the other such image is algorithmically known as well.

---

**Theorem 5.** Under the inversion of a plane with respect to a circle  $c(O, r)$  any point  $A \neq P_\infty$  that is located outside of  $c$  is transformed into an image point  $A' \neq O$  that is located inside of  $c$ .

**Proof.** For any such point  $A$  that is located outside of  $c$ , according to **Condition 3** of **Definition 1**, we have:

$$|OA| \times |OA'| = r^2$$

Because the point  $A \neq P_\infty$  is located outside of the circle of inversion, it is the case that  $|OA| > r$ . Hence:

$$|OA'| = \frac{r^2}{|OA|} < \frac{r^2}{r} = r$$

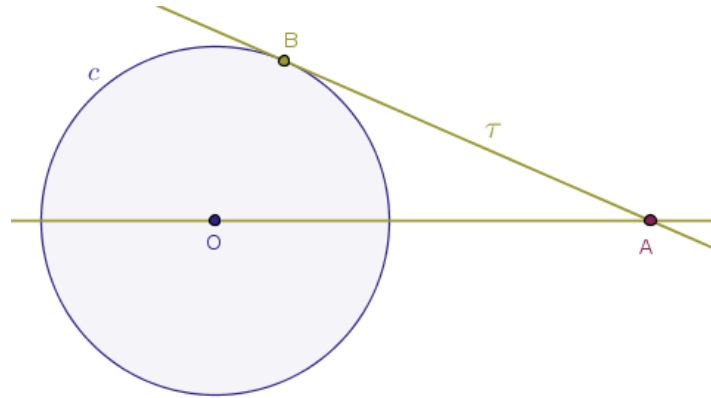
from where  $|OA'| < r$  and, since the original point  $A$  is distinct from the point at infinity  $P_\infty$ , the conclusion of the theorem follows.  $\square$

Note again that the above theorem applies to inversions with both negative and positive powers alike.

**Problem 3.** Given a circle of inversion  $c(O, r)$  and a point  $A \neq P_\infty$  that is located outside of  $c$ , construct  $A$ 's inverse with respect to  $c$  with positive power.

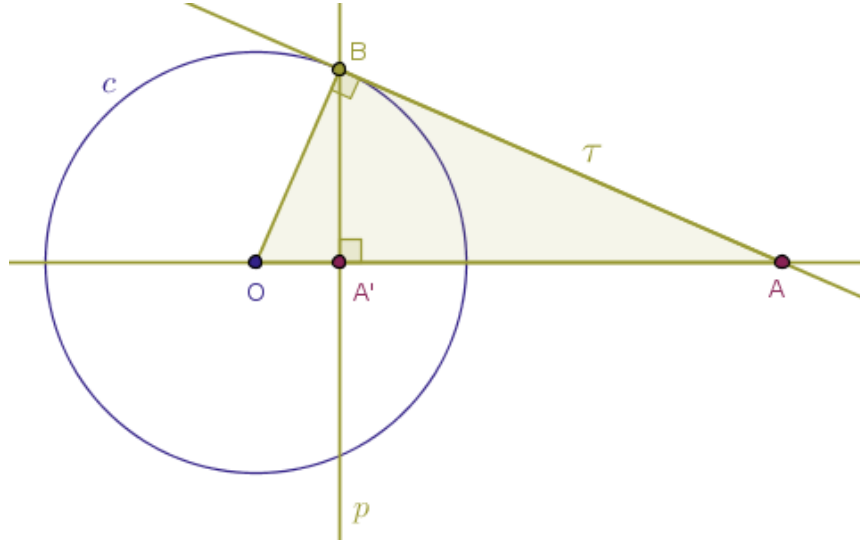
**Solution.** Our definition of inversion of a plane with respect to a circle with positive power suggests that one way to solve this problem is to reverse the steps of **Problem 1**, as follows.

1. Construct a straight line through the points  $O$  and  $A$ , draw one straight line,  $\tau$ , tangent to the circle of inversion  $c$  through the point  $A$ . Let  $\tau$  touch  $c$  at the point  $B$  (Figure 3.9):



A tangent to a circle through a point located outside of that circle can be constructed with Euclidean tools alone in a number of different ways. In this book we will take it that all such ways are already known to us.

2. Draw a straight line,  $p$ , through the point  $B$  perpendicular to the straight line  $OA$  until  $p$  intersects  $OA$  at the point  $A'$  (Figure 3.10):



The point  $A'$  thus obtained is the sought-after image of the original point  $A$  under the inversion of the parent plane with respect to the circle  $c$  with positive power.

For a proof observe that, again, the right, by construction, triangles  $OBA$  and  $OA'B$  are similar and in such triangles the lengths of the sides about the shared interior angle at the shared vertex  $O$  are in the same proportion:

$$\frac{|OA'|}{|OB|} = \frac{|OB|}{|OA|}$$

from where:

$$|OA| \times |OA'| = |OB|^2 = r^2$$

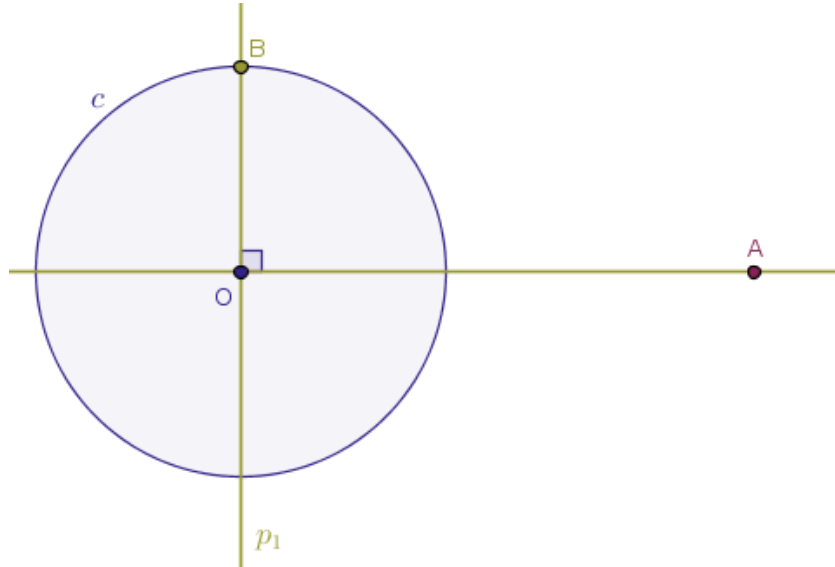
which, along with the usual collinearity of the points  $O$ ,  $A$  and  $A'$ , validates the claim.  $\square$

---

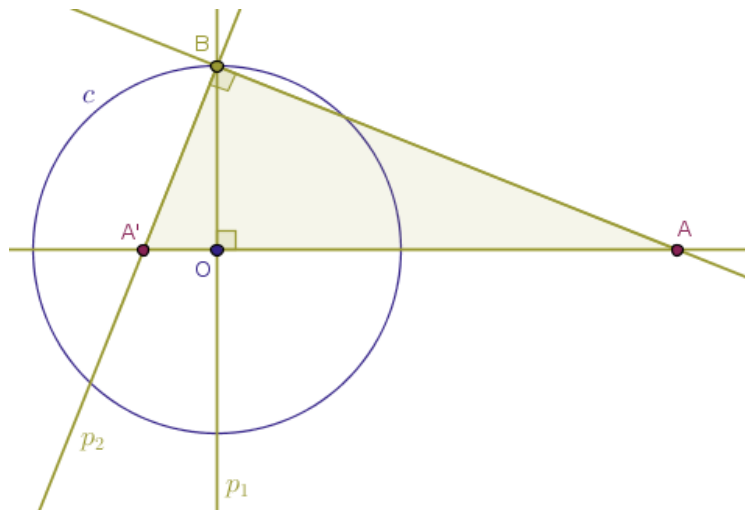
**Problem 4.** Given a circle of inversion  $c(O, r)$  and a point  $A \neq P_\infty$  that is located outside of  $c$ , construct  $A$ 's inverse with respect to  $c$  with negative power.

**Solution.** Our definition of inversion of a plane with respect to a circle with negative power suggests that one way to solve this problem is to reverse the steps of **Problem 2**, as follows.

1. Construct a straight line through the points  $O$  and  $A$ , draw a straight line,  $p_1$ , through the point  $O$  perpendicular to the straight line  $OA$  until  $p_1$  intersects  $c$  at the point  $B$  (Figure 3.11):



2. Draw a straight line through the points  $A$  and  $B$ , draw a straight line,  $p_2$ , through the point  $B$  perpendicular to the straight line  $AB$  until  $p_2$  intersects the straight line  $OA$  extended at the point  $A'$  (Figure 3.12):



The point  $A'$  thus obtained is the sought-after image of the original point  $A$  under the inversion of the parent plane with respect to the circle  $c$  with *negative* power.

Proof. The points  $A$ ,  $O$  and  $A'$  are collinear and coplanar with the plane containing the circle of inversion  $c$  by construction. Also by construction, since the angle  $ABA'$  is right, the center of inversion,  $O$ , is located in between the image points  $A$  and  $A'$ .

The triangle  $ABA'$  is right and the straight line  $BO$  that emanates from the triangle's right-angle vertex  $B$  is perpendicular to the triangle's base  $AA'$  also by construction. Hence, by Euclid's "*Elements*" Book 6 Proposition 8, the triangles  $AOB$  and  $A'OB$  are similar. But in similar triangles, by the same Book 6 Proposition 4, the lengths of the following sides about the equal angles of these triangles are in the same proportion:

$$\frac{|OA|}{|OB|} = \frac{|OB|}{|OA'|}$$

from where:

$$|OA| \times |OA'| = |OB|^2 = r^2$$

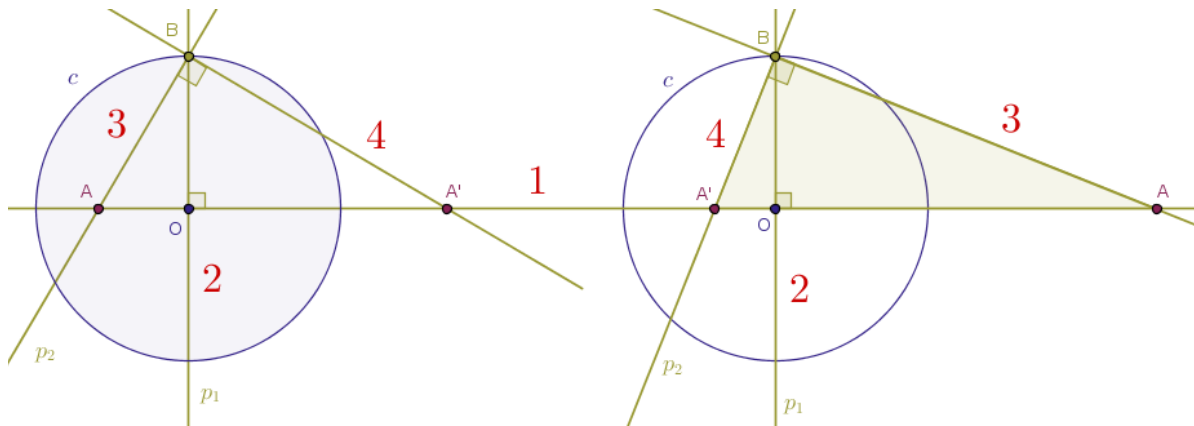
which satisfies **Condition 3** of **Definition 1**.  $\square$

### Constructive Symmetry On Parade

If we go back to the construction steps that we generated in the solutions of **Problem 2** and **Problem 4** then we will notice that these steps are symmetrical in the following sense.

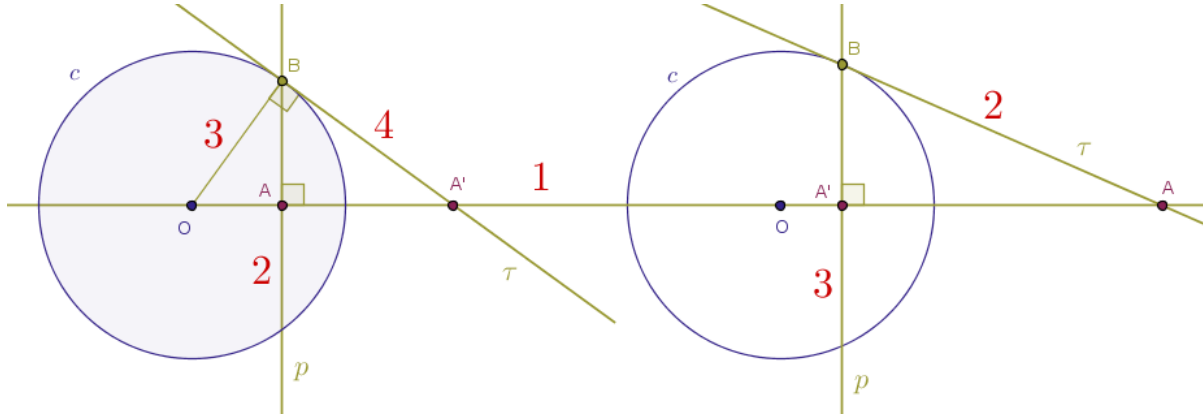
The exact location of the preimage point,  $A$ , with respect to the circle of inversion, **inside** or **outside** of it, does not affect the sequence of the said steps when the power of inversion is negative.

In order to see that, let us number the construction steps of Problem 2 and Problem 4 and put them side by side (Figure 3.13):



In other words, we see that when the power of inversion is negative then the description of the algorithm according to which the respective image is constructed remains one and the same or symmetric with respect to the location of the original point relative to the circle of inversion.

However, when the power of inversion is positive then the steps that we have uncovered in **Problem 1** and **Problem 3** are not symmetric because when the preimage point is located **inside** of the circle of inversion, as shown on the left of the diagram below, then in order to locate the image of that point with positive power we first construct a perpendicular to the straight line  $OA$  and only then we construct a tangent to the circle of inversion (Figure 3.14):



But when the original point is located **outside** of the circle of inversion, as shown on the right in Figure 3.14 above, then in order to locate the image of that point with positive power we first construct a tangent to the circle of inversion and only then we draw a perpendicular to the straight line  $OA$  through the point of tangency,  $B$ .

Hence, the goal of the next problem is to erase the above non-symmetry in the construction steps of the inverse image of a point with positive power and invent an algorithm that is completely symmetric in the sense described above.

---

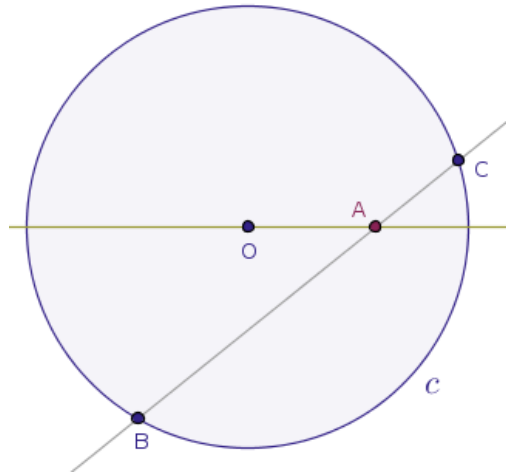
**Problem 5.** Given a circle of inversion  $c(O, r)$ , find a symmetric way to construct the image  $A'$  of a given point  $A \neq O$  and  $A \neq P_\infty$  under the inversion of the parent plane with respect to  $c$  with positive power, a way that works the same regardless of whether the preimage point is located in or outside of the circle of inversion.

**Solution.** At an intuitive level in this problem we want to invent a way to construct two similar triangles that have the radius of the circle of inversion as one of their sides and that have the distances  $|OA|$  and  $|OA'|$  as their other sides, because then through a corresponding proportion we will derive the relation that satisfies **Condition 3** of **Definition 1**.

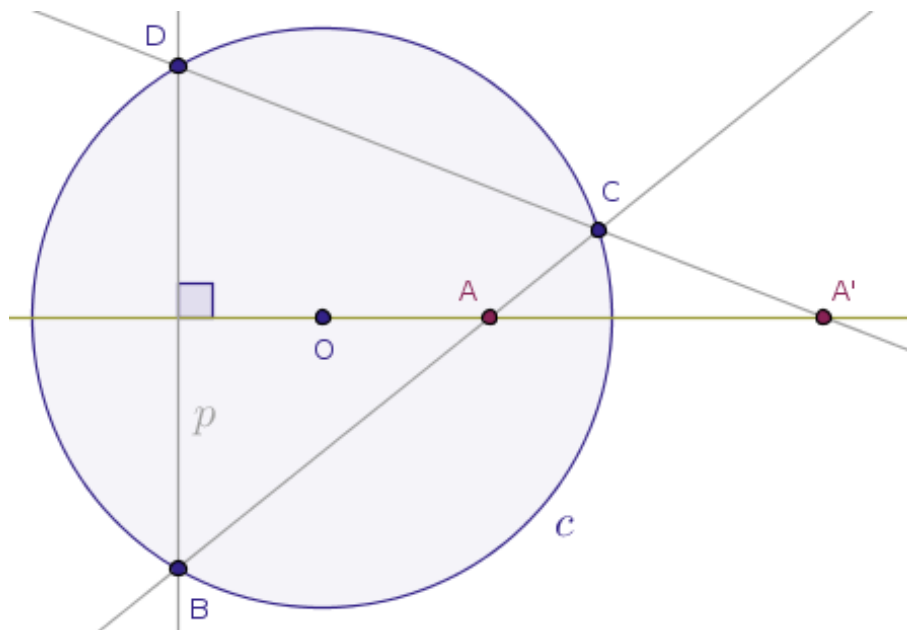
The above two similar triangles cannot be haphazard and have to be positioned advantageously with respect to the circle of inversion. Finding these two triangles is the challenge of this problem and here is one way to meet that challenge.

Assume that the original point  $A$  is located **inside** of the circle of inversion. Then, in order to locate the inverse image of the point  $A$  with positive power in a symmetric way, follow these steps.

1. Draw a straight line through the points  $O$  and  $A$ , draw an arbitrary secant of the circle  $c$  that intersects  $c$  at two distinct points,  $B$  and  $C$  (Figure 3.15):

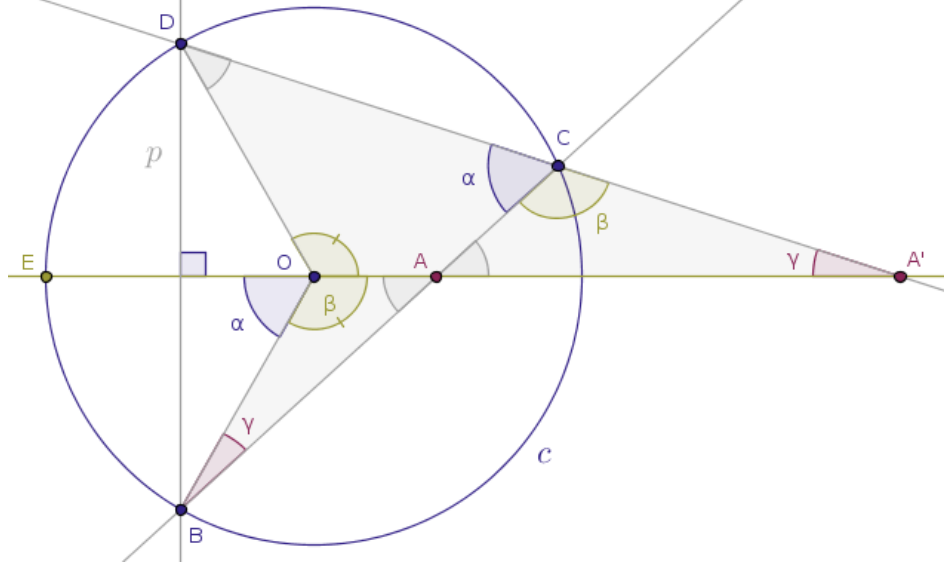


2. Draw a straight line,  $p$ , through the point  $B$  perpendicular to the straight line  $OA$  until  $p$  intersects the circle  $c$  at the point  $D$ , draw a straight line  $DC$  until it intersects the straight line  $OA$  at the point  $A'$  (Figure 3.16):



The point  $A'$  thus constructed is the image of the point  $A$  sought-after.

Proof. Let the straight line  $OA$  intersect the circle of inversion  $c$  at the point  $E$  that lies outside of the line segment  $AA'$  and let the above points  $B$  and  $D$  be the reflections of each other in the straight line  $OA$  (Figure 3.17):



Then, the angle  $DOB$  is central and, by Euclid's "*Elements*" Book 3 Proposition 20, the magnitude of that angle is twice the magnitude of the angle  $DCB$  that subtends the same circular arc  $DEB$ :  $\angle DOB = 2 \cdot \angle DCB$ .

By construction, since the triangle  $DOB$  is isosceles, with  $|OB| = |OD| = r$ , and the straight line  $OE$  is the bisector of its interior angle at the vertex  $O$ , by magnitude the angle  $EOB$  is also one half of the magnitude of the angle  $DOB$ :  $\angle DOB = 2 \cdot \angle EOB$ .

Hence,  $\angle EOB = \angle DCB = \alpha$ . Next, the angles  $EOA$  and  $DCA'$  are straight:

$$\angle EOA = \angle DCA' = \pi$$

and, since by magnitude all straight angles are equal one another and the angles  $BOA$  and  $ACA'$  are supplementary:

$$\angle EOA = \angle EOB + \angle BOA$$

and:

$$\angle DCA' = \angle DCB + \angle BCA'$$

it follows that these supplementary angles are equal by magnitude as well:

$$\angle BOA = \angle ACA'$$



since the points  $B$  and  $A$  are collinear and are located on the same side of the straight line  $DC$ .

The magnitudes of the interior angles of the triangles  $BOA$  and  $ACA'$  at the vertex  $A$  are equal one another:

$$\angle BAO = \angle CAA'$$

because these angles are vertical. Hence, the remaining interior angles in these triangles must be equal by magnitude as well:

$$\angle ABO = \angle AA'C = \gamma$$

The triangles  $DOA'$  and  $ACA'$  share the same vertex  $A'$  in such a way that the magnitudes of their interior angles at that vertex are one and the same:

$$\angle DA'O = \gamma = \angle CA'A$$

Lastly, since the triangle  $BOD$  is isosceles and  $OE$  is the bisector of its interior angle at  $O$ , it follows that the magnitudes of the interior angles at  $O$  of the triangles  $BOA$  and  $DOA'$  must be equal one another as well:

$$\angle BOA = \angle DOA'$$

which, in turn, means that the triangles  $BOA$  and  $DOA'$  are similar by **AAA**.

In such similar triangles the lengths of their sides about the equal angles are in the same proportion where the respective sides lie opposite the equal angles:

$$\frac{|OA|}{|OB|} = \frac{|OD|}{|OA'|}$$

from where:

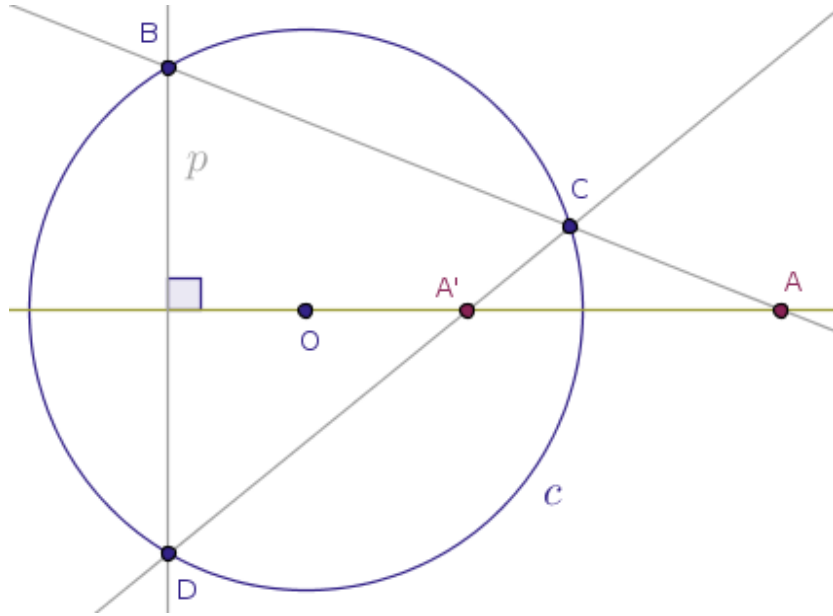
$$|OA| \times |OA'| = |OB| \times |OD| = r \times r = r^2$$

Thus, since all the conditions of **Definition 1** hold, it follows that yes, indeed, the point  $A'$  thus constructed is the inverse image of the point  $A$  with respect to the circle  $c$  with positive power.

In order to see how the above construction steps will work when the point  $A$  is located outside of the circle of inversion, we simply put the point  $A$  outside of that circle,  $c$ , but we still follow the same steps as they are laid out above.

That is.

When the original point  $A$  jumps **outside** of the circle of inversion then we see that even though that point trades places with its image point  $A'$  and so do the points  $B$  and  $D$  (Figure 3.18):



the description of the steps that we gave earlier *does not change* and *remains one and the same*. Which precisely means that our algorithm described above is symmetric.  $\square$

As a useful exercise, our readers are encouraged to generate a proof of the converse statement that we proved in **Problem 5** on their own.

**Proposed theorem.** If the points  $A$  and  $A'$  are the inverses of each other with respect to a circle  $c$  with positive power then the above points  $B$  and  $D$  are the reflections of each other in the straight line  $AA'$ .

**Proof.** Exercise. Hint, Euclid's "*Elements*" Book 6 Proposition 5 may be handy.

From the theorems that we proved and the problems that we solved in this chapter we already see that it is plausible that:

*if a point  $A'$  is the inverse of a point  $A$  under the inversion of a plane with respect to a circle with either power, positive or negative, then the point  $A$  is the inverse of the point  $A'$  of the same plane with respect to the same circle with the same power*

We will prove that theorem officially in the next chapter.

But, coupled with the fact that inversions of a plane in one and the same circle with one and the same power, positive or negative, are equivalent in the sense that we explained above, these two features of inversion mean that we can exploit them to our advantage by choosing the most convenient inversion

type and/or point to derive a proof or a solution of a problem and then claiming the same for the remaining inversion type and/or point.